

Day 31 - IPV6 (pt 1)

Sunday, September 6, 2026

10:46 AM

IPv6 Information

IPv6

IPv6 is a protocol that was introduced for a very important reason: A lack of IPv4 addresses.

- An **IPv6** address is **128 bits**
- Every additional **bit** doubles the number of possible addresses. For that reason, **IPv6** contains more than enough addresses and will probably never run out.
- Every **IPv6** address is represented with **(8) quartets**.
 - Ex: **2001 : 0DB8 : 5917 : EABD : 6562 : 17EA : C92D : 59BD**
 - Every **quartet** consists of **(4) hexadecimal** digits, and is separated with a colon (:)
 - Each **hexadecimal** digit is equivalent to **(4) binary** bits OR **(2) decimal** numbers

Hexadecimal Review

There are three numbering systems you need to know for the CCNA:

1. **Binary** / Base **2** / **0b**
 - 0, 1
2. **Decimal** / Base **10** / **0d**
 - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
3. **Hexadecimal** / Base **16** / **0x**
 - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Binary to Hexadecimal Conversion

There are 4 steps needed for a binary to hexadecimal conversion:

1. Split the number into **4-bit** groups
2. Convert each **4-bit** group to **decimal**
3. Convert each **decimal** number to **hexadecimal**
4. Assemble the **hexadecimal** digits together

Example of binary to hexadecimal conversion

Let's say we have binary bits '0b11011011' and we want to convert it to hexadecimal digits. We'd follow this process:

1. **Step 1** (Split the number into **4-bit** groups)
 - **0b1101 0b1011**
2. **Step 2** (Convert each **4-bit** group to **decimal**)
 - **0d13 0d11**
3. **Step 3** (Convert each **decimal** number to **hexadecimal**)
 - **0xD 0xB**
4. **Step 4** (Assemble the **hexadecimal** digits together)
 - Answer = **0xDB**

Hexadecimal to Binary Conversion

The hexadecimal to binary conversion is basically the reversed version of bin to hex, utilizing 4 steps as well:

1. Split the **hexadecimal** digits
2. Convert each **hexadecimal** digit to **decimal**
3. Convert each **decimal** number to **binary**
4. Assemble the **binary** bits together

Example of hexadecimal to binary conversion

Let's say we have hexadecimal digits '0xDB' and we want to convert it to binary bits. We'd follow this

Let's say we have hexadecimal digits 0x2B and we want to convert it to binary bits. We follow this process:

1. **Step 1** (Split the **hexadecimal** digits)
 - 0x2 0xB
2. **Step 2** (Convert each **hexadecimal** digit to **decimal**)
 - 0d2 0d11
3. **Step 3** (Convert each **decimal** number to **binary**)
 - 0b0010 0b1011
4. **Step 4** (Assemble the **binary** bits together)
 - Answer = 0b00101011

Shortening/Abbreviating IPv6 Addresses

There are two steps in shortening an IPv6 address:

1. **Leading 0's** can be removed.
 - Example:
 - 2001 : 0DB8 : 000A : 001B : 20A1 : 0020 : 0080 : 34BD
 - Will turn into:
 - 2001 : DB8 : A : 1B : 20A1 : 20 : 80 : 34BD
2. **Consecutive quartets of all 0's** can be replaced with a double colon (::)
 - Example:
 - 2001 : 0DB8 : 0000 : 0000 : 0000 : 0000 : 0080 : 34BD
 - Will turn into:
 - 2001 : 0DB8 :: 0080 : 34BD

Note on shortening consecutive quartets of all 0's

Consecutive quartets of all 0's can only be shortened ONCE in an IPv6 address. If we did it twice in the same address, we wouldn't know how many *quartets* of zeros are contained in each double colon.

Expanding Shortened IPv6 Addresses

To expand an IPv6, you just do the shortening steps in reverse:

1. Put **leading 0's** where needed (all *quartets* should have **4 hexadecimal** digits)
 - Example:
 - FE80 : 2 : 0 : 0 : FBE8 : 08BA : CAFE : 08DB
 - Will turn into:
 - FE80 : 0002 : 0000 : 0000 : FBE8 : 08BA : CAFE : 08DB
2. If a double colon (::) is used, replace it with all-0 *quartets*. Make sure to have **8 quartets** total.
 - Example:
 - FE80 :: 0002 : 0000 : 0000 : FBE8
 - Will turn into:
 - FE80 : 0000 : 0000 : 0000 : 0002 : 0000 : 0000 : FBE8

Finding the IPv6 Prefix (in Global Unicast Addresses)

- Typically, ISPs will give enterprises **/48 IPv6** blocks.
- However, **IPv6 subnets** use a **/64** prefix length
- That means that enterprises have **16 bits** to make *subnets* with.
- The remaining **64 bits** are used for hosts.

Global Unicast Address

A global unicast is a public IPv6 address. To help explain what it is, here's an example global unicast address:

- **2001 : 0DB8 : 8B00 : 0001 : 0000 : 0000 : 0001 /64**
- The **global unicast** address consists of 3 different sections:
 1. **Global Routing Prefix (48 bits)**
 - A prefix that's assigned by the ISP

- Its location in the previous example: '2001 : 0DB8 : 8B00'
- 2. **Subnet Identifier (16 bits)**
 - Used to make various subnets
 - Its location in the previous example: '0001'
- 3. **Interface Identifier (64 bits)**
 - Represents the host portion of the address
 - Its location in the previous example: '0000 : 0000 : 0000 : 0001'

Finding the Prefix

I had trouble explaining this section of IPv6 over text, and I believe it is easier to understand with the visual representations. I recommend that you go and check out the Jeremy's IT Lab lecture video for a clearer explanation.

- In **IPv6** addresses, every **hexadecimal** digit represents **4 binary** bits, therefore, every **quartet** represents **16 binary** bits. Also, **prefix lengths** represent the number of **bits** that are fixed in the IP address.
 - If you're asked to find the **prefix length** for an **IPv6** address, you'd start counting **hexadecimal** digits from the left until you've reached the **hexadecimal** digit that contains the specified **prefix length**. Afterwards, to find the network address, you set all the **host bits** to **zero** and that's it.
- However, the difficulty of this task can change depending on what **prefix length** you're being asked for.
 - If you're asked to find the **prefix length** and that number is a multiplicative of **4**, then you can just do the previously explained steps.
 - If you're asked to find the **prefix length** and the number isn't a multiplicative of **4**, then you'll have to find the **hexadecimal** digit that contains the **binary** bit we're looking for.

Examples of finding the prefix:

1. Finding the prefix for **/56**
 - In this example, we kept counting **hexadecimal** digits until we found the one that contains the **56th** bit, that digit being **1**.
 - And then we set all the **host bits** on the right to **zero**.

Finding the IPv6 prefix

1632485256

300D:00F2:0B34:2100:0000:0000:1200:0001/56

↓

300D:F2:B34:2100:::/56

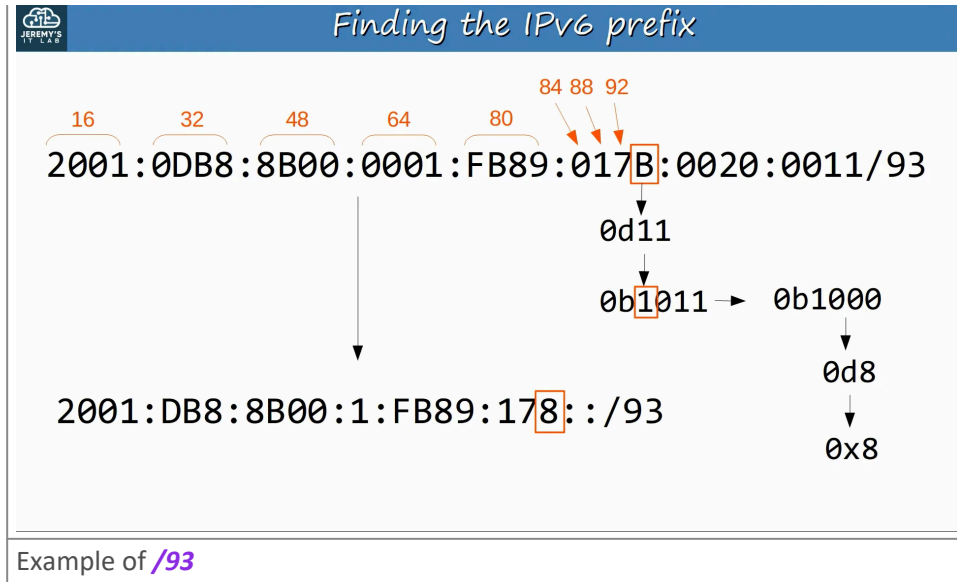
300D:F2:B34:21::/56

=

300D:00F2:0B34:0021::/56

Example of **/56**

2. Finding the prefix for **/93**
 - In this example, we kept counting **hexadecimal** digits until we found the one that contains the **93rd** bit, that digit being **B**.
 - If we turn the **hexadecimal** digit **B** into **binary** form, we'll get **0b1011**.
 - The first bit in **0b 1011** represents the **93rd** bit we're looking for. So that's where the **prefix length** ends.
 - And then we set all the **host bits** on the right to **zero**



Regional Internet Registries

IPv4 address assignments are controlled by the Internet Assigned Numbers Authority (IANA), which distributes IPv4 address space to Regional Internet Registries (RIR), which then assign them to companies. The five RIRs are:

1. **AFRINIC**
 - Controls *Africa*
2. **APNIC**
 - Controls *Asia-Pacific*
3. **ARIN**
 - Controls *Canada*, many *Caribbean* and *North Atlantic* islands, and the *US*
4. **LACNIC**
 - Controls *Latin America* and the *Caribbean*
5. **RIPE NCC**
 - Controls *Europe*, the *Middle East*, and parts of *Central Asia*